

VPIMA801 - Project description
MSc in Quantum Computing

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This project investigates the application of hybrid quantum-classical optimisation to resource allocation in Cell-Free Massive MIMO communication networks. In these systems, multiple distributed access points cooperate to serve a large number of users, making the joint assignment of access points and communication time slots a challenging combinatorial optimisation problem [Özlem Tuğfe Demir(2023)].

The project will first consider a scheduling formulation in which the access-point assignment is fixed and the objective is to determine which users can be served simultaneously. This problem will be represented as a conflict graph and solved through graph colouring techniques by repeatedly identifying independent sets of compatible users. This approach is closely related to the hybrid quantum-classical framework for graph colouring and Maximum Independent Set developed by Coelho et al [W. da Silva Coelho and Henry(2023)].

The independent-set subproblem will be formulated as a Quadratic Unconstrained Binary Optimisation (QUBO) problem, following established formulations for graph optimisation problems [W. da Silva Coelho and Henry(2023), Lucas(2014)]. The project will investigate alternative graph representations that jointly encode scheduling and access point assignment. Rather than assuming a single graph formulation, several representations will be explored, where graph nodes may represent users, access points, or time slots, while graph edges describe the relationships between the remaining communication resources. The goal is to identify graph models that accurately capture the communication problem while remaining well suited to quantum optimisation.

A central component of the project will be the development of a hybrid quantum-classical optimisation workflow. The classical part will generate realistic communication scenarios, construct the graph representations, formulate the optimisation problem, and evaluate the resulting schedules. The computationally demanding graph optimisation subproblems will then be solved using quantum optimisation methods executed on Pasqal's¹ emulator and, if available, on neutral atom quantum hardware. The quantum and classical components will be integrated into a complete optimisation framework, allowing the influence of different graph formulations on both communication performance and quantum algorithm performance to be evaluated.

The communication scenarios will build upon an existing Cell-Free Massive MIMO simulator developed during previous work on access point assignment. The main contributions of the project will include the design of multiple graph representations, the development of corresponding hybrid quantum-classical optimisation methods, their implementation and integration into a complete workflow, and an experimental comparison of the proposed formulations from both

¹The project includes a few short research visits at Pasqal's facilities in France to facilitate collaboration and experimental work.

the communication and quantum computing perspectives.

References

- [Lucas(2014)] A. Lucas. Ising formulations of many np problems. *Frontiers in Physics*, 2014.
- [W. da Silva Coelho and Henry(2023)] L. Henriët W. da Silva Coelho and L.-P. Henry. A quantum pricing-based column generation framework for hard combinatorial problems. *arXiv*, 2023.
- [Özlem Tuğfe Demir(2023)] Luca Sanguinetti Özlem Tuğfe Demir, Emil Björnson. *Foundations of User-Centric Cell-Free Massive MIMO*. 2023.